

Chapter 21

#Electric current $i = \frac{dq}{dt}$

#Coulomb's law $F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2}$

Chapter 22

#Definition of Electric field $\vec{E} = \frac{\vec{F}}{q_0}$

#Field Due to a Point Charge

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$$

#Field Due to an Electric Dipole

$$E = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3}$$

#Field Due to Charged

disk $E = \frac{\sigma}{2\epsilon_0} (1 - \frac{z}{\sqrt{z^2 + R^2}})$

#Force on a Point charge in an Electric

Field $\vec{F} = q\vec{E}$

#Dipole in an Electric Field

$$\vec{\tau} = \vec{p} \times \vec{E}, U = -\vec{p} \cdot \vec{E}$$

Chapter 23

#Gauss' law

$$\epsilon_0 \Phi = q_{enc}, \Phi = \oint \vec{E} \cdot d\vec{A}$$

#conducting surface $E = \frac{\sigma}{\epsilon_0}$

#line of charge $E = \frac{\lambda}{2\pi\epsilon_0 r}$

#sheet of charge $E = \frac{\sigma}{2\epsilon_0}$

#spherical shell, for $r > R$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

#spherical shell, for $r < R$ $E = 0$

#A uniform sphere of charge

$$E = (\frac{q}{4\pi\epsilon_0 R^3})r$$

Chapter 24

#Electric potential $V = \frac{-W_\infty}{q_0} = \frac{U}{q_0}$

#Electric potential energy

$$U = qV, \Delta U = q\Delta V = q(V_f - V_i)$$

#Mechanical energy

$$\Delta K = -q\Delta V, \Delta K = -q\Delta V + W_{app}$$

#Finding V from E

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}$$

$$V = -\int_i^f \vec{E} \cdot d\vec{s}, \Delta V = -E\Delta x$$

#Potential Due to a Charged Particle

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}, V = \sum_{i=1}^n V_i = \sum_{i=1}^n \frac{q_i}{r_i}$$

#Potential Due to an Electric Dipole

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \quad (r \gg d)$$

#Potential Due to a Continuous Charge

Distribution $V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$

#Calculating E from V

$$E_s = -\frac{\partial V}{\partial s}, E = -\frac{\Delta V}{\Delta s}$$

#Potential energy for two particles

$$U = W = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

Chapter 25

#Definition of Capacitance

$$q = CV$$

#A parallel-plate capacitor $C = \frac{\epsilon_0 A}{d}$

#A cylindrical

$$\text{capacitor } C = 2\pi\epsilon_0 \frac{L}{\ln(b/a)}$$

#A spherical

capacitor $C = 4\pi\epsilon_0 \frac{ab}{b-a}$

#An isolated sphere $C = 4\pi\epsilon_0 R$

#capacitors in parallel $C_{eq} = \sum_{j=1}^n C_j$

#capacitors in series $\frac{1}{C_{eq}} = \sum_{j=1}^n \frac{1}{C_j}$

#Potential energy $U = \frac{q^2}{2C} = \frac{1}{2} CV^2$

#Energy density $u = \frac{1}{2} \epsilon_0 E^2$

#Gauss' Law with a Dielectric

$$\epsilon_0 \oint \kappa \vec{E} \cdot d\vec{A} = q$$

Chapter 26

#Electric current $i = \frac{dq}{dt}$

#Current density $i = \int \vec{J} \cdot d\vec{A}$

#Drift speed $\vec{J} = (ne)\vec{v}_d$

#Definition of R $R = \frac{V}{i}$

#Definitions of ρ and σ

$$\rho = \frac{1}{\sigma} = \frac{E}{J}$$

$\vec{E} = \rho \vec{J}, R = \rho \frac{L}{A}$

#Change of ρ with temperature

$$\rho - \rho_0 = \rho_0 \alpha (T - T_0)$$

#Resistivity of a metal $\rho = \frac{m}{e^2 n \tau}$

#Power $P = iV$

#Resistivity dissipation $P = i^2 R = \frac{V^2}{R}$



Chapter 27

Emf $\mathcal{E} = \frac{d\Phi}{dt}$

Single loop circuit $i = \frac{\mathcal{E}}{R + r}$

Power

$P = iV$, $P_r = i^2 r$, $P_{emf} = i\mathcal{E}$

Resistances in series $R_{eq} = \sum_{j=1}^n R_j$

Resistance in parallel $\frac{1}{R_{eq}} = \sum_{j=1}^n \frac{1}{R_j}$

Charging a capacitor, $RC = \tau$

$q = C\mathcal{E}(1 - e^{-t/RC})$

$i = \frac{dq}{dt} = \left(\frac{\mathcal{E}}{R}\right)e^{-t/RC}$

Discharging a capacitor, $RC = \tau$

$q = q_0 e^{-t/RC}$

$i = \frac{dq}{dt} = -\left(\frac{q_0}{RC}\right)e^{-t/RC}$

Chapter 28

Magnetic field $\vec{B}$ $\vec{F}_B = q\vec{v} \times \vec{B}$

A charged particle circulating in a magnetic field

$|q|vB = \frac{mv^2}{r}$, $r = \frac{mv}{qB}$

$f = \frac{\omega}{2\pi} = \frac{1}{T} = \frac{|q|B}{2\pi m}$

The Hall Effect

$n = \frac{Bi}{Vle}$, $V = vBd$

Magnetic force

$\vec{F}_B = i\vec{L} \times \vec{B}$, $d\vec{F}_B = id\vec{L} \times \vec{B}$

Magnetic dipole moment

$\mu = NiA$, $\vec{\tau} = \vec{\mu} \times \vec{B}$

Orientation energy of a Magnetic

dipole $U(\theta) = -\vec{\mu} \cdot \vec{B}$

Work done on dipole by the agent

$W_a = \Delta U = U_f - U_i$

Chapter 29

The Biot-Savart Law

$d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{s} \times \hat{r}}{r^2}$

A long straight wire $B = \frac{\mu_0 I}{2\pi r}$

A circular Arc $B = \frac{\mu_0 I \phi}{4\pi R}$

Force between parallel currents

$F_{ba} = I_b L B_a \sin 90^\circ = \frac{\mu_0 L I_a I_b}{2\pi d}$

Ampere's Law

$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$

Field of an ideal solenoid $B = \mu_0 nI$

Field of an ideal toroid $B = \frac{\mu_0 nI}{2\pi r}$

Field of a magnetic dipole (circular current-carrying coil)

$\vec{B}(z) = \frac{\mu_0}{2\pi} \frac{\vec{\mu}}{z^3}$

Constants

Speed of light $c = 3.00 \times 10^8 \text{ m/s}$

Elementary charge

$e = 1.60 \times 10^{-19} \text{ C}$

Free-fall acceleration $g = 9.81 \text{ m/s}^2$

Gravitational constant

$G = 6.67 \times 10^{-11} \text{ m}^3/\text{s}^2 \cdot \text{kg}$

Permittivity constant

$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$

$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$

Permeability constant

$\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A} = 1.26 \times 10^{-6} \text{ T} \cdot \text{m/A}$

Electron mass

$m_e = 9.109 \times 10^{-31} \text{ kg}$

Proton mass $m_p = 1.673 \times 10^{-27} \text{ kg}$

Neutron mass

$m_n = 1.675 \times 10^{-27} \text{ kg}$

Prefix for SI Units

Factor	Prefix	Symbol
10^{15}	peta-	P
10^{12}	tera-	T
10^9	giga-	G
10^6	mega-	M
10^3	kilo-	K
10^2	hecto-	h
10^1	deka-	da
10^{-1}	deci-	d
10^{-2}	centi-	c
10^{-3}	milli-	m
10^{-6}	micro-	μ
10^{-9}	nano-	n
10^{-12}	pico-	p
10^{-15}	femto-	f

Integrals

$\int \frac{1}{x} dx = \ln|x| + C$

$\int e^{-ax} dx = -\frac{1}{a} e^{-ax} + C$

$\int x^m dx = \frac{x^{m+1}}{m+1} + C \quad (m \neq -1)$

$\int \sin x dx = -\cos x + C$

#

$\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{x^2 + a^2}) + C$

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$\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2(x^2 + a^2)^{1/2}} + C$

